

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (12 Marks)

Suppose, you have built a circuit as shown in **Figure 1** in the hardware lab. In the lab, you also have two bulbs at your disposal. The first bulb, '**Bulb-1**' has 40Ω resistance and the second bulb, '**Bulb-2**' has unknown resistance but different than the '**Bulb-1**'. Answer the following questions:

- i) **Determine** the Thevenin equivalent circuit for the circuit you have built in the lab.
- ii) If you connect **Bulb-1** between **a-b**, then **determine** how much power will this bulb get?
- iii) If you connect **Bulb-2**, you get exactly the same power as the **Bulb-1** gets. Then, **determine** the resistance of **Bulb-2**.
- iv) Do you get maximum power at the **a-b** for **Bulb-1**? If not, your teachers suggested to make changes in the original circuit so that **Bulb-1** get maximum power. **Explain** what changes you should make as per your teacher's suggestion to the original circuit so that **Bulb-1** achieves maximum power.

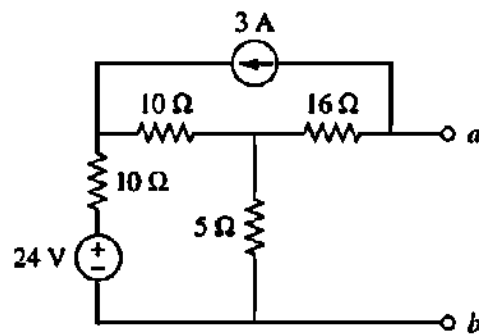


Figure 1

Question 2: Answer all the questions. (8 Marks)

Answer the following questions for circuit shown in **Figure 2**:

- i) Draw the circuits turning off the independent sources.
- ii) From these circuits, determine the current, **i** using Superposition theorem.

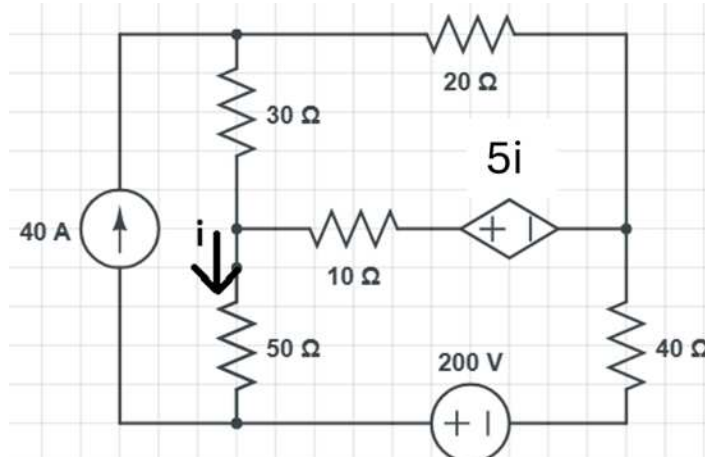


Figure 2

Question 3: Answer all the questions

(12 Marks)

In 'Figure 3', the current source, $i_s(t) = 8\cos(\omega t + 30^\circ)$ A. Now, answer the following questions if $\omega = 200\text{Krad/s}$: [2+2+8]

- Draw the phasor-domain circuit.
- Determine the equivalent impedance of this circuit.
- Find the expressions for $v(t)$, $i_1(t)$, $i_2(t)$ and $i_3(t)$.

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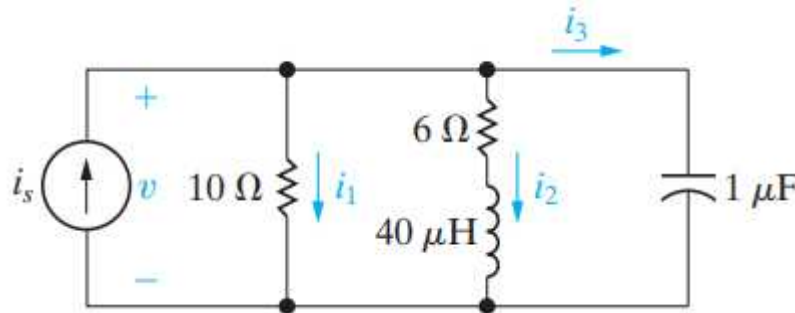


Figure 3

Question 4: Answer all the questions.

(8 Marks)

- Find the RMS value, I_{RMS} of the current waveform shown in Figure 4(a).
- Use the RMS value from (i) as the maximum amplitude of the sinusoidal current source in the circuit shown in Figure 4(b). Determine V_x and the average real power absorbed by the 20-ohm resistor, assuming an angular frequency of 100 rad/s.

[4+4]

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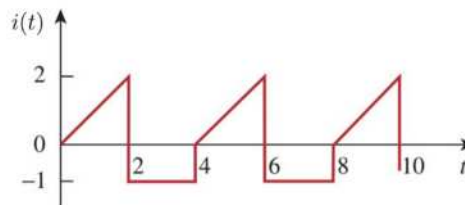


Figure 4(a)

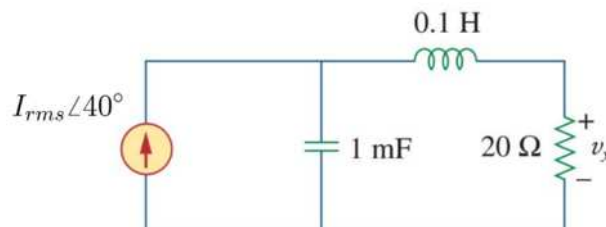


Figure 4(b)